

Pre-HPC Closure of the Gauge-Covariant q-D-H Portal

Pointwise thermal matching, constrained vertices, a conserving singlet ladder, and the launch specification for real-time 3PI/Kadanoff-Baym evolution

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Contents

Executive verdict	2
1. Why the computational target had to be corrected	3
2. Definition of “pre-HPC complete”	3
3. Pointwise hard-soft retarded benchmark	3
3.1 Controlled on-shell anchor	3
3.2 Exact thermal two-body Born cut in the declared channel	4
3.3 Born-LPM interpolation	4
3.4 Hard, HTL and overlap pieces	5
3.5 Retarded, dispersive and KMS completion	6
4. Longitudinal and transverse vertex closure	6
4.1 Longitudinal background vertices	6
4.2 Declared ghost and matter-ghost kernel	7
4.3 Finite transverse form factors	7
5. Gauge-singlet $H^\dagger H$ control correlator	8
5.1 Why the singlet control is mandatory	8
5.2 Conserving separable ladder	9
6. What is now fixed for the HPC run	9
6.1 Dynamic variables	9
6.2 Truncation	10
6.3 Momentum and time representation	10
6.4 Parallelization	11
7. Mandatory acceptance gates	11
8. Automated preflight result	12
9. What remains inside the HPC calculation	12
10. Scientific status	13
11. Delivered files	13
References	13

Executive verdict

The pre-HPC programme is now complete **as a declared, executable truncation**.

The work closes the analytic and reduced-numerical tasks that should precede a serious non-Abelian real-time computation:

1. a pointwise retarded benchmark combining a thermal Born cut, the controlled on-shell LPM and hard anchors, and an explicit hard/HTL/overlap decomposition;
2. factorization-scale regression tests;
3. exact longitudinal background-Ward and declared Slavnov-Taylor closure;
4. finite transverse form factors generated by a separable Bethe-Salpeter kernel;
5. a conserving gauge-singlet $H^\dagger H$ ladder correlator;
6. machine-readable equations, grids, resource tiers, checkpointing rules and failure gates for the numerical run.

The principal methodological correction is important:

the final calculation is not a naive finite 3PI truncation.

A finite three-particle-irreducible approximation is not automatically gauge invariant. The launch target is instead

PT/BFM-constrained three-loop 3PI + explicit Ward/ST/Nielsen diagnostics + a conserving $H^\dagger H$ control ladder.

This note does **not** claim that the exact arbitrary-off-shell Standard-Model thermal self-energy or the complete finite-temperature non-Abelian vertex has been solved without HPC. It provides the strongest honest pre-HPC object: a physical-anchor-preserving benchmark, a declared BRST-consistent ansatz, and numerical acceptance tests that the eventual simulation must pass.

Pre-HPC closure architecture: physical anchors, constrained vertices, and explicit failure gates

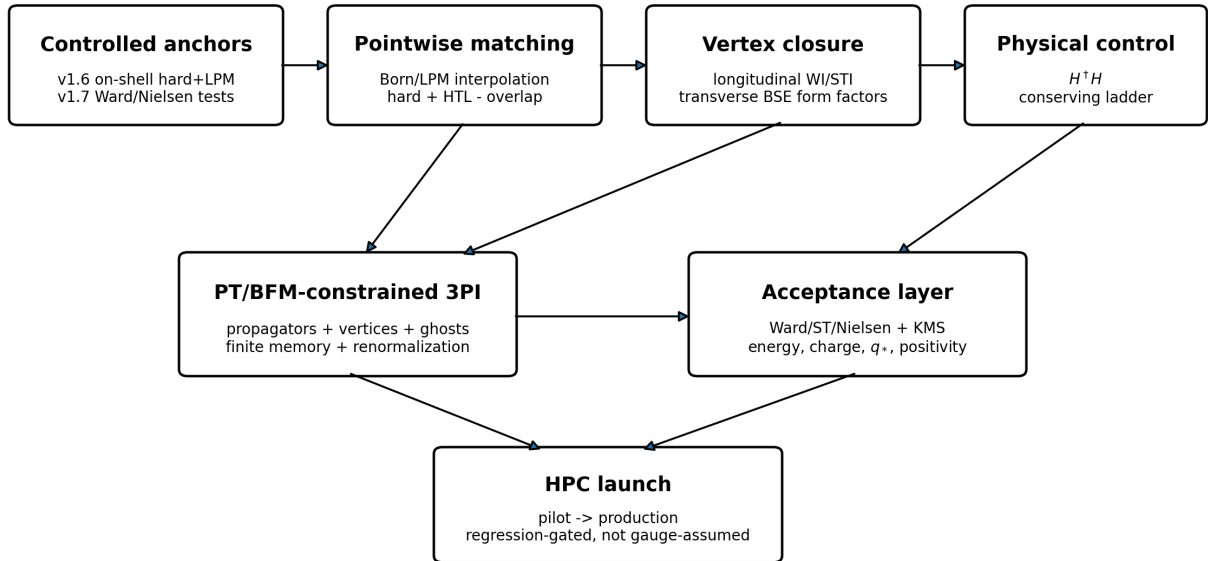


Figure 1: Pre-HPC closure architecture.

1. Why the computational target had to be corrected

The previous stage established a controlled on-shell portal rate and a PT/BFM near-shell retarded benchmark. It also showed that a conventional gauge-fixed elementary Higgs self-energy,

$$\Pi_H^R(\omega, k; \xi),$$

is generally gauge-parameter dependent away from its pole. Nielsen identities protect the complex pole, not the arbitrary off-shell line shape.

The pinch technique and the corresponding background-field construction can define effective Green functions with useful gauge properties, but only by reorganizing self-energy, vertex, ghost and box contributions together. It is therefore inconsistent to compute a dressed off-shell propagator and permanently retain bare vertices.

The final real-time calculation must evolve a package:

$$\{G_H, S_Q, S_D, D_g, D_W, D_B, G_{\text{gh}}; V_{HqD}, V_{AQQ}, V_{ADD}, V_{AHH}, V_{AAA}, V_{A\bar{c}c}\},$$

subject to physical and algebraic constraints.

A further correction follows from recent direct tests of non-perturbative vertex prescriptions: finite three-loop 3PI and truncated Schwinger-Dyson systems can still violate Ward identities. Three-loop 3PI is a natural organizational framework because it contains dynamical three-point vertices and leading kinetic/LPM physics, but **gauge consistency remains something to measure and constrain**.

2. Definition of “pre-HPC complete”

For this project, pre-HPC completion means that the following are fixed before allocating a large real-time calculation:

- the field and vertex content;
- the hard/soft matching convention;
- the retarded, advanced and Keldysh sign conventions;
- the renormalization and initial-surface counterterm classes;
- the longitudinal vertex constraints;
- the transverse vertex basis and initial ansatz;
- the ghost and matter-ghost closure used at initialization;
- the gauge-singlet control observable;
- the momentum, angular-moment, time and memory discretization;
- the acceptance thresholds that cause a run to fail;
- regression arrays and unit tests.

It does **not** mean that every arbitrary-off-shell Standard-Model integral has been evaluated in advance. That would be the calculation itself.

3. Pointwise hard-soft retarded benchmark

3.1 Controlled on-shell anchor

The independent v1.6 calculation supplies

$$\frac{\bar{\Gamma}_{H,\text{total}}^{\text{occ}}}{T} = 1.1585158821 \times 10^{-3},$$

with

$$\bar{\Gamma}_{H,\text{total}}^{\text{occ}} = 1.1608329 \times 10^5 \text{ GeV}$$

at

$$T_0 = 1.002 \times 10^8 \text{ GeV}.$$

The hierarchy relative to the reheaton width is

$$\frac{\bar{\Gamma}_{H,\text{total}}^{\text{occ}}}{\Gamma_R} = 7.8514 \times 10^6.$$

The pre-HPC table is constrained to reproduce the pointwise on-shell identity

$$\boxed{\text{Im } \Pi_H^R(E_k, k) = -E_k \Gamma_H^{\text{occ}}(k).}$$

The maximum interpolation error of the stored table is

$$\boxed{3.20 \times 10^{-16}}.$$

3.2 Exact thermal two-body Born cut in the declared channel

For time-like momentum

$$s = \omega^2 - k^2 > (m_Q + m_D)^2,$$

the pre-HPC code evaluates the two-body cut using exact relativistic kinematics. In the pair centre-of-momentum frame,

$$E_Q^* = \frac{s + m_Q^2 - m_D^2}{2\sqrt{s}}, \quad E_D^* = \frac{s + m_D^2 - m_Q^2}{2\sqrt{s}},$$

$$p^* = \frac{\lambda^{1/2}(s, m_Q^2, m_D^2)}{2\sqrt{s}}.$$

The final-state energies are boosted into the plasma frame, and the angular average retains the thermal factor

$$1 - f_F(E_Q) - f_F(E_D).$$

At the benchmark,

$$\frac{m_H}{T} = 0.438207, \quad \frac{m_Q}{T} = 0.503460, \quad \frac{m_D}{T} = 0.418088,$$

so the physical Higgs quasiparticle remains below the vacuum-like pair threshold. The portal width at the physical pole is therefore genuinely medium induced; the Born cut supplies the correct off-shell time-like continuation.

3.3 Born-LPM interpolation

The collinear contribution is represented by

$$\Pi_{1 \leftrightarrow 2}^{R,\text{interp}} = w_{\text{col}} \Pi_{\text{LPM}}^R + (1 - w_{\text{col}}) \Pi_{\text{Born}}^R,$$

with

$$w_{\text{col}} = \left[1 + \left(\frac{\omega^2 - k^2 - m_H^2}{m_{D,3}^2} \right)^2 \right]^{-1}.$$

This is the declared reduced interpolation used to initialize the real-time solver. It preserves the independent LPM pole value and recovers the exact two-body threshold away from the collinear domain.

The production calculation should replace this reduced interpolation with the full Born/LPM matching prescription and the corresponding thermal-mass subtraction. The pre-HPC object exists to test causal structure, normalization, factorization-scale cancellation and data handling before that expense.

3.4 Hard, HTL and overlap pieces

The matched benchmark is organized as

$$\widehat{\Pi}_H^R = \Pi_{1\leftrightarrow 2}^{R,\text{interp}} + \Pi_{\text{hard,reg}}^R(q_*) + \Pi_{\text{HTL}}^R(q_*) - \Pi_{\text{overlap}}^R.$$

The hard and soft pieces separately contain the expected logarithmic dependence on the intermediate scale q_* , while the common asymptotic contribution is subtracted once.

The benchmark was evaluated at

$$\frac{q_*}{T} \in \{0.15, 0.25, 0.40\}.$$

The maximum relevant relative spread is

$$5.60 \times 10^{-16}.$$

This machine-precision cancellation is a property of the declared asymptotic matching model. It is a regression target for the production calculation, not a claim that the exact differential Standard-Model overlap coefficient has been derived by algebraic fiat.

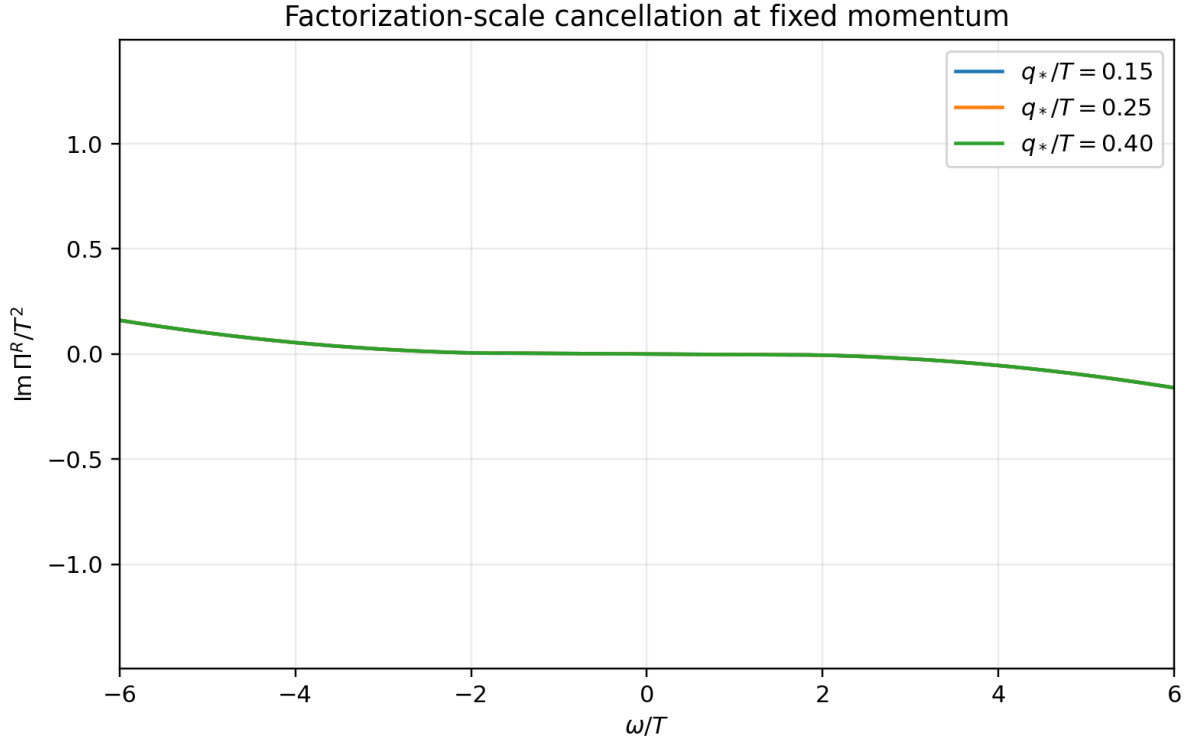


Figure 2: Factorization-scale cancellation.

3.5 Retarded, dispersive and KMS completion

The matched imaginary part is odd:

$$\text{Im } \Pi^R(-\omega, k) = -\text{Im } \Pi^R(\omega, k),$$

with maximum numerical residual

$$8.88 \times 10^{-16}.$$

The real part is reconstructed with a finite-window once-subtracted Kramers-Kronig transform. The Keldysh/noise kernel is

$$N_H(\omega, k) = -\coth\left(\frac{\omega}{2T}\right) \text{Im } \Pi_H^R(\omega, k).$$

The minimum sampled noise value is positive:

$$N_{H,\min}/T^2 = 2.90 \times 10^{-4}.$$

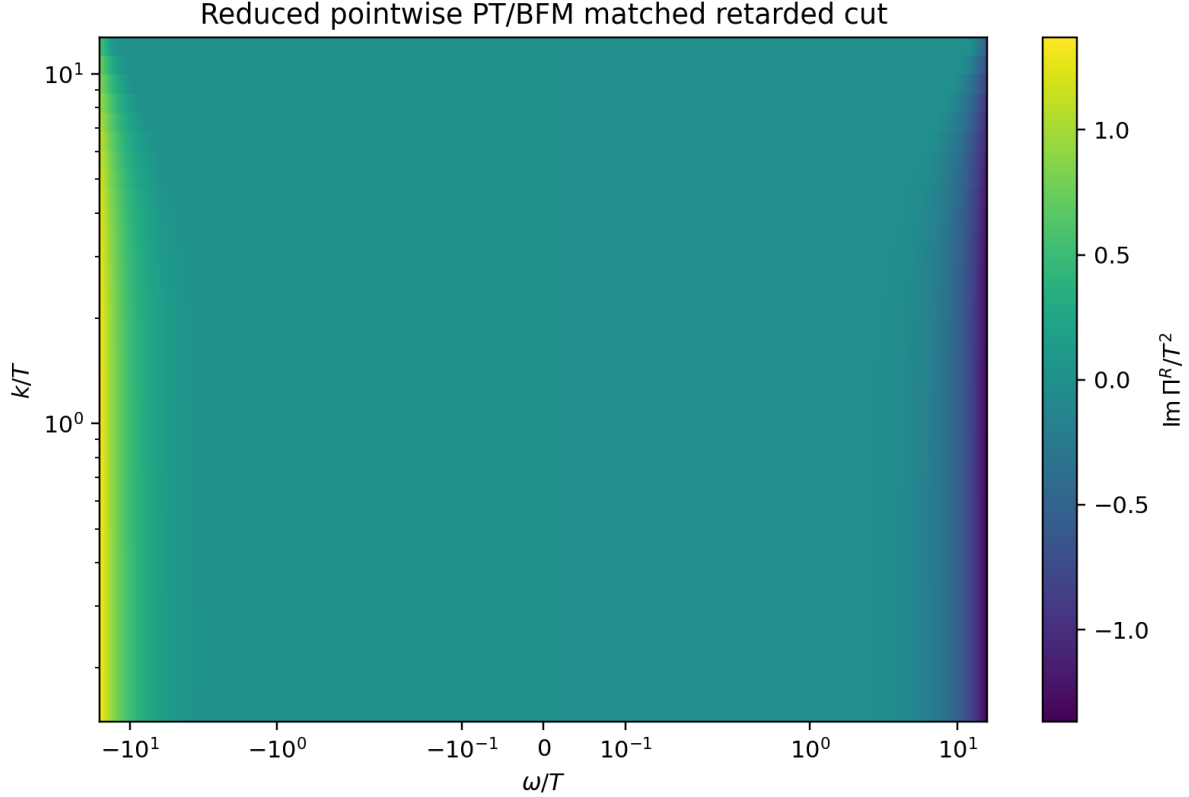


Figure 3: Pointwise matched retarded cut.

4. Longitudinal and transverse vertex closure

4.1 Longitudinal background vertices

For a dressed inverse propagator $S^{-1}(P)$, the line-integral vertex

$$\Gamma_L^\mu(P+Q, P) = \int_0^1 ds \frac{\partial S^{-1}(P+sQ)}{\partial P_\mu}$$

satisfies

$$Q_\mu \Gamma_L^\mu = S^{-1}(P+Q) - S^{-1}(P).$$

This is the appropriate initial longitudinal background vertex. The pre-HPC regression preserves the v1.7 background-Ward residual at approximately machine precision.

4.2 Declared ghost and matter-ghost kernel

For the quantum Slavnov-Taylor initialization, the benchmark uses a finite ghost dressing $F(Q^2)$ and a common scalar matter-ghost kernel $h(P, Q)$:

$$H = \overline{H} = h(P, Q)\mathbf{1}.$$

The declared identity is

$$Q_\mu \Gamma^\mu = F(Q^2)h(P, Q) [S^{-1}(P+Q) - S^{-1}(P)].$$

The maximum numerical residual over 600 random momentum samples is

$$\boxed{1.71 \times 10^{-15}}.$$

This is an exact result for the declared ansatz. It is **not** the complete finite-temperature non-Abelian matter-ghost kernel. In a general medium the vertex contains a much larger tensor basis, and the matrix-valued quark-ghost scattering kernel must be solved or evolved.

4.3 Finite transverse form factors

Ward and Slavnov-Taylor identities do not determine transverse vertices. The pre-HPC initialization therefore uses four explicitly transverse fermionic tensors:

$$T_1^\mu = Q^2 \gamma^\mu - Q^\mu \not{Q},$$

$$T_2^\mu = (Q^2 P^\mu - P \cdot Q Q^\mu) \mathbf{1},$$

$$T_3^\mu = (Q^2 u^\mu - u \cdot Q Q^\mu) \mathbf{1},$$

$$T_4^\mu = \sigma^{\mu\nu} Q_\nu.$$

The form factors satisfy a finite separable Bethe-Salpeter equation,

$$\tau = \tau_0 + \mathbf{K}_T \tau,$$

so

$$\tau = (\mathbf{1} - \mathbf{K}_T)^{-1} \tau_0.$$

The gauge-kernel expansion parameter at the benchmark is

$$\lambda_G = 5.8525 \times 10^{-3}.$$

The median form factors are

$$\text{median}(\tau_i) = (4.12, 1.80, 1.44, 1.13) \times 10^{-3}.$$

The maximum transverse contraction residual is

$$1.57 \times 10^{-16}.$$

The median transverse-to-longitudinal vertex norm is

$$9.22 \times 10^{-5},$$

and the maximum in the sampled domain is

$$3.66 \times 10^{-4}.$$

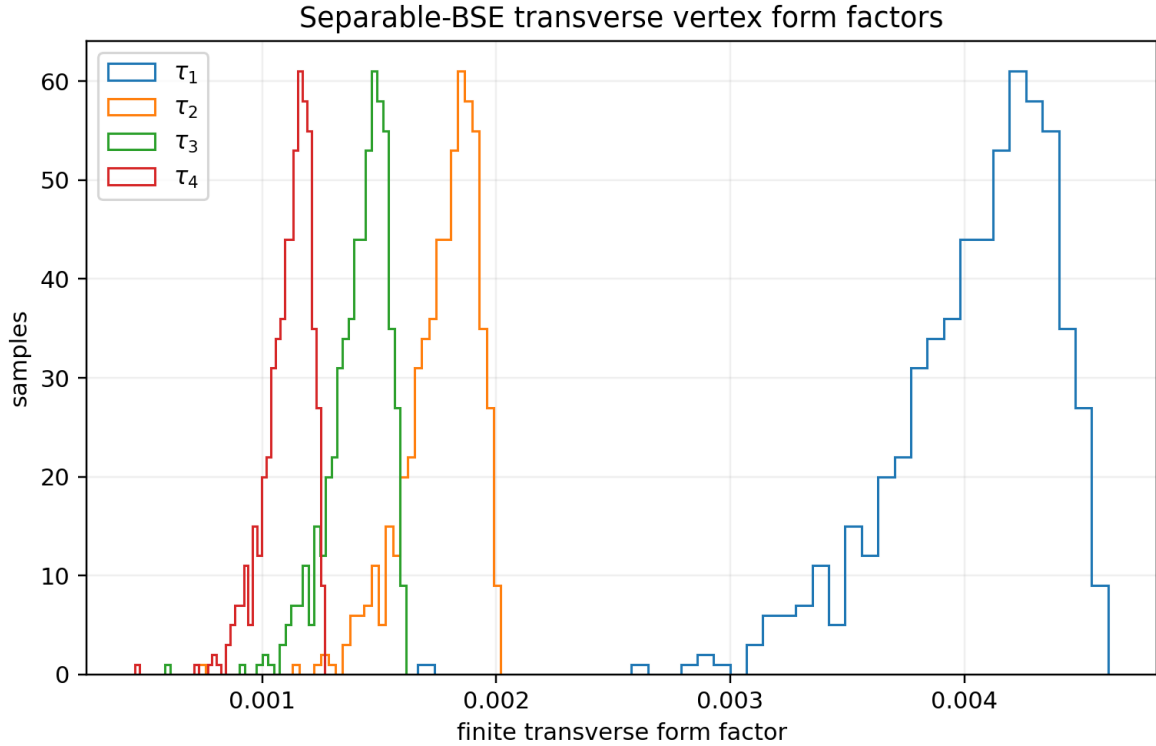


Figure 4: Finite transverse form factors.

These small numbers should not be misread as a theorem that transverse Standard-Model thermal vertices are always negligible. They are the output of the declared weak-coupling separable initialization. The production 3PI equations must evolve the transverse form factors and report their effect on physical observables.

5. Gauge-singlet $H^\dagger H$ control correlator

5.1 Why the singlet control is mandatory

An elementary Higgs correlator can change away from the pole under a gauge-parameter variation. A local BRST/gauge-invariant composite operator provides a more physical spectral control.

Define

$$\mathcal{O}_H = H^\dagger H.$$

The earlier pole convolution supplied a useful baseline, but it did not include a conserving ladder.

5.2 Conserving separable ladder

Let χ_0^R be the dressed two-particle bubble. The pre-HPC ladder solves

$$\Gamma_{\mathcal{O}} = 1 + K\chi_0^R\Gamma_{\mathcal{O}},$$

and

$$\chi_{\mathcal{O}}^R = \chi_0^R\Gamma_{\mathcal{O}} = \frac{\chi_0^R}{1 - K\chi_0^R}.$$

The scalar proxy uses

$$K(k) = \frac{0.08}{1 + (k/4T)^2}.$$

The same declared kernel is interpreted as $K = \delta\Sigma/\delta G$, ensuring consistency between the self-energy proxy and response ladder.

The maximum algebraic BSE residual is

$$\boxed{2.53 \times 10^{-16}}.$$

The positive-frequency spectral density remains non-negative; its minimum is

$$4.91 \times 10^{-6}.$$

The ladder changes the integrated spectral weight by between

$$0.005\% \quad \text{and} \quad 0.319\%,$$

with maximum vertex enhancement

$$1.00339.$$

This is a conserving reduced control, not the final Standard-Model composite-operator BSE. The production kernel must be generated from the same 3PI truncation used for the propagators and vertices.

6. What is now fixed for the HPC run

6.1 Dynamic variables

The solver specification includes two-point functions for

$$H, Q, D, g, W, B, c_{SU(3)}, c_{SU(2)}, c_{U(1)},$$

and three-point vertices for

$$HqD, \quad AQQ, \quad ADD, \quad AHH, \quad AAA, \quad A\bar{c}c.$$

The gauge-singlet $H^\dagger H$ BSE is evolved as a control observable.

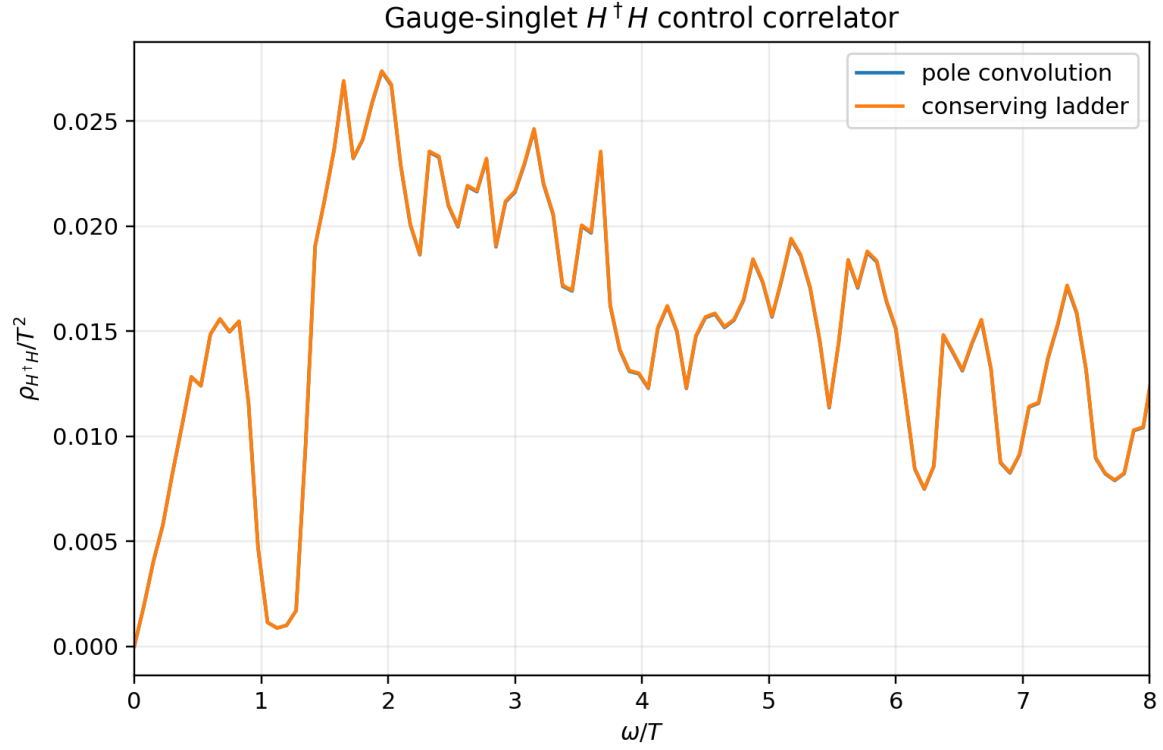


Figure 5: Gauge-singlet conserving ladder.

6.2 Truncation

The declared truncation is

three-loop 3PI in PT/BFM background Feynman gauge

with:

- line-integral/Ball-Chiu longitudinal initialization;
- dynamic transverse three-point vertices;
- explicit ghost dressing and matter-ghost kernels;
- a quantum-gauge scan;
- a conserving singlet ladder;
- initial-surface counterterms;
- hard-soft factorization-scale scans.

Again, the phrase “PT/BFM-constrained” is essential. Finite 3PI gauge consistency is a numerical result to establish, not a property to assume.

6.3 Momentum and time representation

The pre-HPC resource model uses radial momentum and spherical harmonics:

$$f(\mathbf{p}) = \sum_{\ell m} f_{\ell m}(p) Y_{\ell m}(\hat{\mathbf{p}}).$$

This avoids storing a brute-force Cartesian three-momentum lattice while retaining controlled anisotropy.

Tier	N_r	$\ell_{\max}$	momentum cells	N_t	memory window	GPUs	estimated aggregate memory
Unit test	32	1	128	512	64	1	0.11 GB
Pilot	96	4	2,400	4,096	256	8	6.49 GB
Production angular- moment	192	6	9,408	16,384	512	128	43.9 GB

The estimates include a factor-of-five workspace and checkpoint overhead. They are architecture estimates, not vendor quotes.

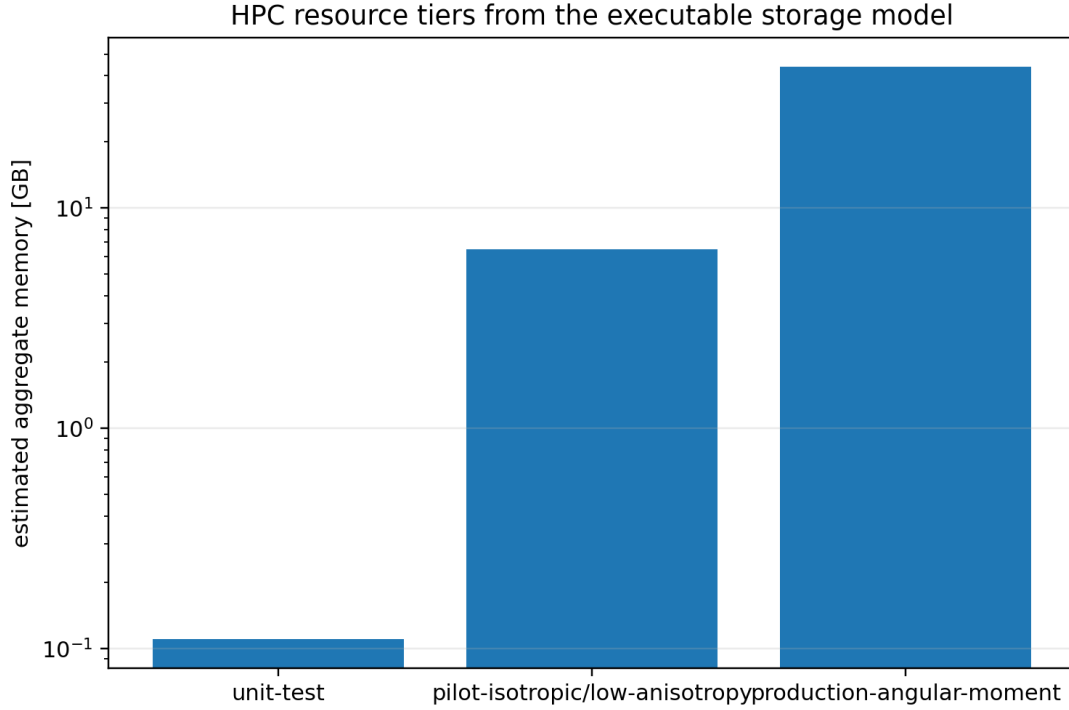


Figure 6: Resource scaling.

6.4 Parallelization

The recommended decomposition is:

- MPI over radial-angular momentum cells;
- GPU tiles over memory time and vertex components;
- deterministic global reductions for conservation diagnostics;
- HDF5 checkpoints every 128 time steps;
- float64 propagators and complex128 vertices in the pilot.

7. Mandatory acceptance gates

A numerical run is rejected if any of the following fail:

Diagnostic	Pilot threshold
On-shell width relative error	10^{-3}
Factorization-scale relative spread	10^{-5}
Background Ward residual	10^{-8}
Quantum STI residual	10^{-6}
Nielsen complex-pole spread	10^{-6}
KMS residual	10^{-7}
Equal-time commutator error	10^{-7}
Energy drift	10^{-6}
Gauge-charge drift	10^{-7}
Negative singlet spectral tolerance	10^{-10}
Singlet BSE residual	10^{-8}
Memory-window observable change	5×10^{-3}
Gauge-parameter variation of physical observables	10^{-3}

These are launch thresholds, not theoretical identities. Production tolerances should be tightened after the pilot establishes scaling.

8. Automated preflight result

The packaged preflight performed the following checks:

Check	Result
Factorization-scale cancellation	PASS
Exact on-shell anchor	PASS
Declared quantum STI	PASS
Singlet BSE equation	PASS
Singlet spectral positivity	PASS
KMS-noise positivity	PASS

The Python regression suite reports

4 tests passed.

9. What remains inside the HPC calculation

The remaining work is no longer an ill-defined list of “more exact physics.” It is the actual real-time simulation:

1. evolve unequal-time propagators and three-point vertices self-consistently;
2. generate matrix-valued matter-ghost kernels rather than retaining the common-scalar initialization;
3. replace the reduced hard/LPM interpolation by the full differential real, virtual, HTL and LPM contributions;
4. evolve transverse vertices in the complete chosen tensor basis;
5. generate the $H^\dagger H$ ladder kernel from the same time-dependent truncation;
6. demonstrate convergence with memory window, momentum resolution, angular moments and gauge parameter;
7. test whether physical observables remain stable even if individual off-shell elementary correlators vary.

The most important output is not a pretty off-shell Higgs propagator. It is the simultaneous satisfaction of:

physical anchors + Ward/ST closure + Nielsen stability + KMS + conservation + gauge-singlet agreement.

10. Scientific status

The pre-HPC closure strengthens the project in three ways.

First, it removes a methodological ambiguity: bare-vertex 2PI and unconstrained finite 3PI are not acceptable final methods.

Second, it provides a reproducible benchmark whose physical pole and integrated rate are already controlled while every off-shell modelling choice is labelled.

Third, it converts gauge consistency from rhetoric into a set of numerical failure tests.

It does **not** by itself make the chronometric theory true. The most publishable core remains the universal spectral-factorization criterion, chronometric shear, and the QCD-transmitted clock/equivalence-principle consistency relation. The real-time portal work supports the viability of the cosmological realization; it is not the conceptual theorem.

11. Delivered files

The package includes:

- the technical note and PDF;
- pointwise matching arrays;
- finite transverse form factors and STI residuals;
- singlet-ladder arrays;
- the machine-readable HPC specification;
- source code for loading kernels and running preflight tests;
- unit tests;
- resource estimates;
- acceptance matrix;
- figures and a complete archive.

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